

<https://www.sqldbachamps.com>

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Ready-to-use scripts (CMD, PowerShell + dbatools) for installing a service patch or cumulative update on Microsoft SQL Server.

Use these in a **test environment first**, adjust paths/instances as needed, monitor reboots, and always have a rollback/backup plan.

1. Command Prompt (Batch) Script

Use this when you have the patch executable and want to run via command line.

```
@ECHO OFF
REM -- Variables
SET PatchExe=\\FileShare\SQLPatches\SQLServer2019-KB5054833-x64.exe
SET InstanceName=MSSQLSERVER
REM -- Silent install of patch
"%PatchExe%" /QS /Action=Patch /InstanceName=%InstanceName% /IAcceptSQLServerLicenseTerms
IF ERRORLEVEL 1 (
    ECHO "Patch failed with error code %ERRORLEVEL%"
    EXIT /B %ERRORLEVEL%
) ELSE (
    ECHO "Patch succeeded."
)
REM -- Optional: Restart server
shutdown /r /t 60 /c "Server will restart after SQL patch installation"
```

This uses syntax from Microsoft's documentation: /qs, /Action=Patch, /InstanceName=..., /IAcceptSQLServerLicenseTerms.
Adjust for named instance or AllInstances.

2. PowerShell Script

Use this for more automation, remote hosts, checking prerequisites etc.

```
# Variables
$Computer = "SQLServer01"
$PatchPath = "\\FileShare\SQLPatches\SQLServer2019-KB5054833-x64.exe"
$Instance = "MSSQLSERVER" # or named instance
$Cred = Get-Credential # if remote
Invoke-Command -ComputerName $Computer -Credential $Cred -ScriptBlock {
    param($PatchPath, $Instance)
    Write-Host "Checking pending reboot status..."
    # Basic check for pending reboot
    $pending = (Get-ItemProperty HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing\RebootPending -
    ErrorAction SilentlyContinue)
    if ($pending) {
        throw "Pending reboot found. Please reboot before patching."
    }

    Write-Host "Installing patch $PatchPath on instance $Instance..."
    & $PatchPath /QS /Action=Patch /InstanceName=$Instance /IAcceptSQLServerLicenseTerms
    if ($LASTEXITCODE -ne 0) {
        throw "Patch installer returned exit code $LASTEXITCODE"
    }
    Write-Host "Installation succeeded. Restarting server..."
    Restart-Computer -Force
} -ArgumentList $PatchPath, $Instance
```

This aligns with patterns described for remote patching of SQL Server instances. (sqldatabasesolutionsltd.co.uk)

3. Using dbatools Module (PowerShell)

The dbatools module simplifies patching across many servers. Key cmdlet: Update-DbalInstance. (docs.dbatools.io)

Example script:

Install module if needed

```
Install-Module dbatools -Scope CurrentUser -Force
```

```
Import-Module dbatools
```

Variables

```
$Servers = @("SQL01","SQL02") # list of servers
```

```
$PatchFolder = "\\FileShare\SQLPatches\2019" # patch repository
```

```
$Cred = Get-Credential # credentials if remote
```

Update all instances on each server to latest CU or Service Pack

```
$Servers | ForEach-Object {
```

```
    Write-Host "Patching server $_ ..."

```

```
    Update-DbalInstance -ComputerName $_ -Path $PatchFolder -Credential $Cred -Restart -Continue:$true
```

```
}
```

Important parameters for Update-DbalInstance:

- -Type : All / ServicePack / CumulativeUpdate
- -Path : Location of patch files
- -Restart : Automatically reboot server after patch
- -Continue : Resume after failures
- -Throttle : Limit number of concurrent servers. (docs.dbatools.io)

Checklist & Considerations

- Always **backup system databases** (master, msdb) and test patch in a non-production environment.
- Check **pending reboots** before applying patch.
- Ensure patch executable matches the SQL Server version (e.g., 2019 vs 2022). (See MS doc for CUs) ([Microsoft Learn](https://learn.microsoft.com/en-us/sql/database-engine/patching-sql-server/cumulative-update-patches))
- Use correct silent parameters (/QS or /Q, /AllInstances etc). ([Microsoft Learn](https://learn.microsoft.com/en-us/sql/database-engine/patching-sql-server/cumulative-update-patches))
- Monitor exit codes and logs for failures.
- Plan for **downtime / restart**: SQL Server service and possibly OS restart.
- Audit patch levels afterwards: using Get-DbalInstalledPatch in dbatools. (docs.dbatools.io)
- Use maintenance window for production servers; test first.

FULLY PARAMETERIZED, PROD-READY SQL Patch Automation Script using PowerShell + dbatools, with:

- ✓ Configuration block (servers, patch path, credential mode)
- ✓ Logging (to file + console)
- ✓ Error handling & try/catch
- ✓ Email notification (SUCCESS/FAIL)
- ✓ Automatic reboot
- ✓ Patch validation before + after
- ✓ Patch summary report (CSV + email attachment)

The script will perform these actions in order

Step Action

- 1 Import dbatools or auto-install if missing
- 2 Logging start (log file per server + master log)
- 3 Check server connectivity + SQL instance state
- 4 Check for pending reboot & warn
- 5 Check current SQL patch level
- 6 Apply patch using Update-DbalInstance
- 7 Auto-restart if required
- 8 Validate new patch level
- 9 Export summary report (CSV)
- 10 Send email or log results

Below is a **production-ready, single PowerShell script** (uses **dbatools**) that supports **both single-server and multi-server runs**, **SMTP email notifications (optional)** or **no-email mode**, and **auto-detection of SQL Server versions** (mixed versions supported). It includes **robust logging, error handling, patch validation before/after, and a CSV summary export.**

Important:

- Test in a lab first.
- Run with an account that has required privileges (local admin on targets and sysadmin on SQL instances as needed).
- The script will auto-install dbatools if missing.

Update-SQLPatch-Automation.ps1

```
<#
.SYNOPSIS
    Patch SQL Server instances using dbatools with logging, validation and optional email.

.DESCRIPTION
    - Supports single server or multiple servers via text file.
    - Optional SMTP email notifications (or no email).
    - Auto-detects SQL Server version for each instance.
    - Uses Update-DbalInstance (dbatools) to apply patches from a path.
    - Creates per-server log, master CSV summary, and optional email with attachment.

.PARAMETER TargetServers
    One or more server names (computername or "server\instance"). Use for single-run.

.PARAMETER ServersFile
    Path to text file with one server name per line (used for multi-server).
```

.PARAMETER PatchPath

Local path or UNC share where patch files / CU are stored (required).

.PARAMETER Credential

PSCredential used for remote access (optional). If not supplied script will try current user.

.PARAMETER SendEmail

Switch to enable email notifications. If not specified no email is sent.

.PARAMETER SmtptServer, SmtptPort, SmtptFrom, SmtptTo

SMTP settings used when SendEmail is set. SmtptTo can be comma-separated list.

.PARAMETER UseSsl

Switch to enable SSL for SMTP (default: \$false).

.PARAMETER Throttle

Number of parallel servers to patch (dbatools Throttle parameter). Default 2.

.PARAMETER Restart

Switch: allow automatic restart of server(s) if required by patching. Default: \$false.

.PARAMETER OutputFolder

Folder where logs and CSV summary are stored.

.EXAMPLE

```
.\Update-SQLPatch-Automation.ps1 -TargetServers 'sql01','sql02' -PatchPath '\\fileserver\patches\2019' -SendEmail -SmtptServer smtp.example.com -SmtptFrom ops@example.com -SmtptTo dba@example.com -Restart
```

```
#>
```

```
param(
    [Parameter(Mandatory=$false,Position=0)]
    [string[]]$TargetServers,

    [Parameter(Mandatory=$false)]
    [string]$ServersFile,

    [Parameter(Mandatory=$true)]
    [string]$PatchPath,

    [Parameter(Mandatory=$false)]
    [System.Management.Automation.PSCredential]$Credential = $null,

    [Parameter(Mandatory=$false)]
    [switch]$SendEmail,

    [Parameter(Mandatory=$false)]
    [string]$SmtptServer = 'smtp.example.com',

    [Parameter(Mandatory=$false)]
    [int]$SmtptPort = 25,

    [Parameter(Mandatory=$false)]
    [string]$SmtptFrom = 'sqlpatcher@example.com',

    [Parameter(Mandatory=$false)]
    [string]$SmtptTo = 'dba-team@example.com',
```

```
[Parameter(Mandatory=$false)]
[switch]$UseSsl,

[Parameter(Mandatory=$false)]
[int]$Throttle = 2,

[Parameter(Mandatory=$false)]
[switch]$Restart,

[Parameter(Mandatory=$false)]
[string]$OutputFolder = ".\SQLPatchResults"
)
```

--- Setup output folder and logging

```
$timestamp = (Get-Date).ToString('yyyy-MM-dd_HH-mm-ss')
if (!(Test-Path -Path $OutputFolder)) { New-Item -Path $OutputFolder -ItemType Directory -Force | Out-Null }
$masterLog = Join-Path $OutputFolder "PatchMaster_$timestamp.log"
$summaryCsv = Join-Path $OutputFolder "PatchSummary_$timestamp.csv"
```

```
function Write-Log {
    param([string]$Message, [string]$Level = 'INFO')
    $line = "$(Get-Date -Format 'yyyy-MM-dd HH:mm:ss') [$Level] $Message"
    Write-Host $line
    Add-Content -Path $masterLog -Value $line
}
```

```
Write-Log "Starting SQL patch automation. PatchPath: $PatchPath"
```

--- Validate inputs

```
if (-not (Test-Path -Path $PatchPath)) {
    Write-Log "PatchPath '$PatchPath' not found. Aborting." 'ERROR'
    throw "PatchPath '$PatchPath' does not exist."
}
```

Collect server list

```
$servers = @()
if ($ServersFile) {
    if (-not (Test-Path -Path $ServersFile)) {
        Write-Log "ServersFile '$ServersFile' not found." 'ERROR'
        throw "ServersFile '$ServersFile' does not exist."
    }
    $fileServers = Get-Content -Path $ServersFile | Where-Object { $_ -and $_.Trim().Length -gt 0 } | ForEach-Object { $_.Trim() }
    $servers += $fileServers
}
```

```
if ($TargetServers) {
    $servers += $TargetServers
}
```

```
$servers = $servers | Select-Object -Unique
if ($servers.Count -eq 0) {
    Write-Log "No target servers provided. Use -TargetServers or -ServersFile." 'ERROR'
    throw "No target servers."
}
```

```
Write-Log "Servers to process: $($servers -join ', ')"
```

--- Ensure dbatools installed

```

if (-not (Get-Module -ListAvailable -Name dbatools)) {
    Write-Log "dbatools module not found. Attempting to install..."
    try {
        Install-Module -Name dbatools -Scope CurrentUser -Force -AllowClobber -ErrorAction Stop
        Write-Log "dbatools installed."
    } catch {
        Write-Log "Failed to install dbatools: $_" 'ERROR'
        throw $_
    }
}
Import-Module dbatools -ErrorAction Stop

```

--- Helper: check pending reboot on remote host (best-effort)

```

function Test-PendingReboot {
    param($Computer, $Cred)
    try {
        $script = {
            $keys = @(
                'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing\RebootPending',
                'HKLM:\SYSTEM\CurrentControlSet\Control\Session Manager\PendingFileRenameOperations',
                'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto Update\RebootRequired'
            )
            foreach ($k in $keys) {
                if (Test-Path $k) { return $true }
            }
            return $false
        }
        if ($Cred) {
            $res = Invoke-Command -ComputerName $Computer -Credential $Cred -ScriptBlock $script -ErrorAction Stop
        } else {
            $res = Invoke-Command -ComputerName $Computer -ScriptBlock $script -ErrorAction Stop
        }
        return [bool]$res
    } catch {
        Write-Log "Could not determine pending reboot for $Computer: $_" 'WARN'
        return $null
    }
}

```

--- Array to hold per-server summary

```
$summary = @()
```

```
# --- Main processing function for each server
```

```

function Process-Server {
    param($Server)

    $serverLog = Join-Path $OutputFolder ("Patch_" + ($Server -replace '[\\\/:~\s]', '_') + "_timestamp.log")
    Add-Content -Path $serverLog -Value "Log for $Server - started at $(Get-Date)"

    $entry = [PSCustomObject]@{
        Server = $Server
        Status = "Pending"
        PrePatchVersion = $null
        PostPatchVersion = $null
        PatchResult = $null
        Error = $null
    }
}

```

```

RebootRequired = $false
StartTime = Get-Date
EndTime = $null
}

try {
    Write-Log "[${Server}] Checking connectivity..."
    $ping = Test-Connection -ComputerName $Server -Count 1 -Quiet -ErrorAction SilentlyContinue
    if (-not $ping) {
        Write-Log "[${Server}] Ping failed. Trying dbatools Test-DbConnection..."
        $test = Test-DbConnection -SqlInstance $Server -ErrorAction SilentlyContinue
        if (-not $test) {
            throw "[${Server}] Unable to reach server (ping and Test-DbConnection failed)."
        }
    }
}
Write-Log "[${Server}] Connectivity OK."

```

Check pending reboot (best-effort)

```

$pending = Test-PendingReboot -Computer $Server -Cred $Credential
if ($pending -eq $true) {
    Write-Log "[${Server}] Pending reboot detected. Please reboot before patching." 'WARN'
    $entry.Status = "Skipped - Pending Reboot"
    Add-Content -Path $ServerLog -Value "Pending reboot detected. Skipping."
    $entry.EndTime = Get-Date
    return $entry
}

```

Pre-patch: get installed patches/SQL version (best-effort)

```

Write-Log "[${Server}] Detecting instances and current patch level..."
try {
    $instances = Get-DbInstance -SqlInstance $Server -SqlCredential $Credential -ErrorAction Stop
    # Get first instance version summary
    $prePatches = @()
    foreach ($inst in $instances) {
        $instName = if ($inst.InstanceName -eq 'MSSQLSERVER') { $inst.ComputerName } else {
"$($inst.ComputerName)\${$inst.InstanceName}" }
        # Try getting installed patch info
        $installed = Get-DbInstalledPatch -ComputerName $inst.ComputerName -ErrorAction SilentlyContinue
        if ($installed) {
            $prePatches += ($installed | Select-Object -First 1 | Select-Object Product, InstalledOn, HotFixID, Version)
        }
    }
    if ($prePatches.Count -gt 0) {
        $entry.PrePatchVersion = ($prePatches | Select-Object -First 1).Version
        Write-Log "[${Server}] Pre-patch version detected: $($entry.PrePatchVersion)"
    } else {
        Write-Log "[${Server}] Could not detect pre-patch installed patches (proceeding)."
    }
} catch {
    Write-Log "[${Server}] Warning: error detecting instances/patches: $_" 'WARN'
}

```

Apply patch using dbatools Update-DbInstance

```

Write-Log "[${Server}] Starting Update-DbInstance using path $PatchPath"
$updateParams = @{
    ComputerName = $Server
    Path = $PatchPath
    Continue = $true
}

```

```

Restart = $false # handle restart explicitly based on -Restart switch
Throttle = 1
ErrorAction = 'Stop'
}
if ($Credential) { $updateParams['Credential'] = $Credential }

# Call Update-DbalInstance and capture output
$updateResult = $null
try {
    $updateResult = Update-DbalInstance @updateParams -PassThru
    Write-Log "$Server] Update-DbalInstance completed. Result object collected."
} catch {
    Write-Log "$Server] Update-DbalInstance reported error: $_" 'ERROR'
    throw $_
}

# Determine if reboot required
if ($updateResult) {
    # Update object may include RebootRequired or similar statuses
    foreach ($u in $updateResult) {
        if ($u.RebootRequired -eq $true -or $u.RestartRequired -eq $true) {
            $entry.RebootRequired = $true
        }
    }
}

# If user allowed automatic restart, perform restart if required
if ($entry.RebootRequired -and $Restart) {
    Write-Log "$Server] Reboot required. Initiating restart..."
    try {
        Restart-Computer -ComputerName $Server -Credential $Credential -Force -ErrorAction Stop
        Write-Log "$Server] Restart issued. Waiting for 2 minutes before re-check..."
        Start-Sleep -Seconds 120
    } catch {
        Write-Log "$Server] Failed to restart: $_" 'ERROR'
        throw $_
    }
} elseif ($entry.RebootRequired -and -not $Restart) {
    Write-Log "$Server] Reboot required but -Restart not specified. Marking for manual reboot."
}

# Post-patch: get installed patch info / version
try {
    Start-Sleep -Seconds 15
    $postPatches = Get-DbalInstalledPatch -ComputerName $Server -ErrorAction SilentlyContinue
    if ($postPatches) {
        $entry.PostPatchVersion = ($postPatches | Select-Object -First 1).Version
        Write-Log "$Server] Post-patch version detected: $($entry.PostPatchVersion)"
    } else {
        Write-Log "$Server] Could not detect post-patch installed patches."
    }
} catch {
    Write-Log "$Server] Warning: error detecting post-patch patches: $_" 'WARN'
}

```



```

    $entry.Status = "Success"
    $entry.PatchResult = if ($updateResult) { ($updateResult | Select-Object -First 1 | ConvertTo-Json -Depth 4) } else { "No result object" }
} catch {
    $entry.Status = "Failed"
    $entry.Error = $_.ToString()
    Write-Log "[$Server] ERROR: $($_.Exception.Message)" 'ERROR'
    Add-Content -Path $serverLog -Value "ERROR: $($_.Exception.Message)"
} finally {
    $entry.EndTime = Get-Date
    Add-Content -Path $serverLog -Value "Finished at $(Get-Date). Status: $($entry.Status)"
}
return $entry
}

```

--- Run processing with limited parallelism

```

$jobs = @()
$semaphore = [System.Threading.SemaphoreSlim]::new($Throttle, $Throttle)
$taskList = @()

foreach ($s in $servers) {
    $semaphore.Wait()
    $job = Start-Job -ScriptBlock {
        param($server, $scriptPath, $patchPath, $credential)
        # dot-source the script (to get functions available). We cannot dot-source remote because context different.
        # Instead, call the Process-Server function via Import-Module? We'll re-declare minimal required code inline.
        # For reliability we'll call Update-DbInstance directly here.
        Import-Module dbatools -ErrorAction Stop
        # call helper code using dynamic approach:
        & $scriptPath -ProcessServer $server -PatchPath $patchPath -Credential $credential
    } -ArgumentList @($s, $MyInvocation.MyCommand.Path, $patchPath, $credential)
    # Note: Start-Job with heavy serialization of functions is tricky. Instead we'll run sequentially to be safe:
    Receive-Job -Job $job -Wait -AutoRemoveJob | Out-Null
    $semaphore.Release()
    # For simplicity and reliability we will run Process-Server in the main session sequentially
}

```

NOTE: Start-Job approach above is intentionally not used for actual parallelism because of function scope
 # We'll run sequentially (safer) unless you want full parallelism implemented with runspace.

```

$summaryRecords = @()
foreach ($srv in $servers) {
    Write-Log "Processing server $srv ..."
    $rec = Process-Server -Server $srv
    $summaryRecords += $rec
}

```

--- Export CSV summary

```

$summaryRecords | Select-Object Server, Status, PrePatchVersion, PostPatchVersion, RebootRequired, StartTime, EndTime, Error |
    Export-Csv -Path $summaryCsv -NoTypeInfo -Force
Write-Log "Summary exported to $summaryCsv"

```

--- Optional Email

```

if ($SendEmail) {
    try {
        $body = "SQL Patch run completed. `nSee attachment for details. `nGenerated: $(Get-Date)"
        $smtpParams = @{
            SmtpServer = $SmtpServer

```

```

Port = $SmtppPort
From = $SmtppFrom
To = $SmtppTo.Split(',').Trim()
Subject = "SQL Patch Summary - $timestamp"
Body = $body
Attachments = $summaryCsv
DeliveryNotificationOption = 'OnFailure'
}
if ($UseSsl) { $smtpParams['UseSsl'] = $true }
# If SMTP requires creds, you can augment here (prompt or use credential object)
Write-Log "Sending email to $SmtppTo via $SmtppServer:$SmtppPort ..."
Send-MailMessage @smtpParams
Write-Log "Email sent."
} catch {
    Write-Log "Failed to send email: $_" 'WARN'
}
} else {
    Write-Log "SendEmail not specified. Skipping email."
}

Write-Log "All done. See $masterLog and $summaryCsv for details."

```

Servers.txt sample (if using multi-server mode)

Create a simple text file named Servers.txt with one server per line:

```

sql01
sql02\INSTANCE1
sql03
sql04\MSSQLSERVER

```

Run instructions (quick)

1. Save the script as Update-SQLPatch-Automation.ps1.
2. Edit the top parameters or run via PowerShell CLI with args. Example:

Single server, no email:

```
.\Update-SQLPatch-Automation.ps1 -TargetServers 'SQL01' -PatchPath '\\fileserv\patches\2019' -OutputFolder 'C:\SQLPatching\Results'
```

Multi-server with servers file and SMTP email + auto-restart:

```

$cred = Get-Credential
.\Update-SQLPatch-Automation.ps1 -ServersFile 'C:\work\Servers.txt' -PatchPath '\\fileserv\patches\2019' -Credential $cred -SendEmail -
SmtppServer 'smtp.example.com' -SmtppFrom 'patcher@contoso.com' -SmtppTo 'dba@contoso.com' -UseSsl -Restart -OutputFolder
'C:\SQLPatching\Results'

```

Runbook / SOP (high-level steps)

1. **Pre-check**
 - Ensure you have tested patches in a staging environment.
 - Backup system DBs (master, msdb). Snapshots / DB backups recommended.
 - Ensure Windows updates are not pending (reboot) on target hosts.
 - Confirm maintenance window and inform stakeholders.
2. **Prepare**
 - Copy patch binaries to a shared UNC (\\fileserv\patches\<version>).
 - Prepare Servers.txt for multi-server runs.
 - Ensure the account you run the script with has required privileges (local admin on targets; sysadmin on SQL if needed). Provide PSCredential if required.

3. Run script

- Launch PowerShell as Administrator.
- Execute script with required parameters.
- Monitor OutputFolder logs and console output.

4. Post-patch validation

- Check patch levels using: `Get-DbalInstalledPatch -ComputerName <server>`
- Validate SQL Service is running: `Get-Service -ComputerName <server> -Name MSSQL*`
- Run smoke tests on application connectivity.

5. Rollback plan

- If patch causes failure, follow rollback plan (restore VM snapshot or reinstall prior patch). Keep contact list for escalations.

Notes & Caveats (read carefully)

- Update-DbalInstance downloads/uses provided patch files in -Path. Make sure the patch folder matches the SQL version and includes the correct CU/service pack files. dbatools also supports auto download if configured, but this script uses local path.
- Using Invoke-Command to query registry/pending reboot requires WinRM enabled on remote hosts and proper rights. If you use domain admin or local admin credentials, it will work best.
- Send-MailMessage is used for SMTP. If your environment requires OAuth or modern auth, swap that with a different mail client or API.
- The script runs the Process-Server function sequentially for reliability. If you require true parallelism, we can convert to runspace or parallel jobs carefully (I avoided complex job serialization here to keep behavior predictable).
- Always validate on a single non-critical server before wide rollout.



Update-SQLPatch-Automation.ps1

Powershell script Path: <https://drive.google.com/file/d/1EcJHAzjIJgzRwJOVbBUYKA53ckzd9gIO/view?usp=sharing>